

Technological Institute of the Philippines

College of Computer Studies

Deploying a Dynamic Website with Database on Windows Server (Oracle VirtualBox)

AN ACTIVITY SUBMITTED IN PARTIAL FULFILLMENT

OF THE REQUIREMENT OF SYSTEM ADMINISTRATION AND MAINTENDANCE

Submitted By:

JOHN IVAN ROXAS

KATHERINE SUPAN

Submitted To:

JOSELITO SAN JUAN

Date:

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PROJECT DOCUMENTATION

Field	Entry
Student name(s)	John Ivan Roxas and Katherine Supan
Course / Section	CIT 401 - IT41S1
Instructor	Joselito San Juan
Date submitted	October 27, 2026
VM / server name	
Server IP address	

INTRODUCTION

1.1 Purpose

The purpose of this activity is to deploy Salarize on a Windows Server running inside Oracle VirtualBox. Salarize is a secure payroll and employee management system. We first built it for our Information Assurance and Security II course. It lets administrators manage employee records, salaries, and work statuses in one place. It has four roles: Super Admin, Admin, HR Manager, and Auditor. Each role can only do the tasks it is allowed to do.

In that course, we ran Salarize on a development machine. That setup is fine for building and testing. It is not how a real system is hosted. A real system needs a server that stays online, limits who can reach it, and can be recovered when something fails. This activity moves Salarize into that kind of setup.

This document records the whole process. It shows how we prepared the server, installed the web server and the database, deployed the website, and secured the system. It also shows how we tested the system, resolved the problems we met, and planned for backups and maintenance.

We chose Salarize because it is a dynamic website that depends on a database. It also handles sensitive data, such as employee details and salaries. This makes it a particularly relevant case for practicing real system administration, including security. The activity helps us practice setting up services, controlling access, keeping a system running, and planning for recovery. The document is written for our instructor. It is also detailed enough for another administrator to repeat the setup.

1.2 Scope

This activity covers one deployment of Salarize inside a virtual environment. It includes setting up the server, the web server, and the database, and then deploying and testing the website. It also covers the basic security, backup, and maintenance work that keeps the server running after setup.

In Scope:

- **Virtual Machine Setup**: Setting up a Windows Server in Oracle VirtualBox and recording its specs, network mode, and IP settings. We also take a baseline snapshot before major changes.
- **Web Server**: Installing and configuring IIS and the runtime that Salarize needs. We create the site and check that it loads.
- **Database**: Installing the database and creating the Salarize database, its tables, and sample data. We also create a dedicated user with limited privileges.
- **Website Deployment**: Copying the Salarize files to the server and connecting them to the database. We check that the create, read, update, and delete features work for employee records.

- **Server Security**: Applying basic settings such as firewall rules, a non-default administrator password, a password policy, disabled unused accounts, and directory browsing turned off. Salarize's built-in protections, such as role-based access, login lockout, and audit logs, stay in place.
- **Testing**: Running the seven tests in the template, from site loading to database restart after a reboot.
- **Documentation and Upkeep**: Writing a troubleshooting log and a backup and maintenance plan.
- **System Users**: Salarize is for administrative staff only. The Super Admin, Admin, HR Manager, and Auditor roles are the only users, and regular employees do not log in.

Some tasks are not part of this activity. We left them out to keep the project focused on deploying and securing Salarize on one server. The items below are not covered in this document.

Out of Scope:

- **Public Hosting**: Hosting on a public cloud or a live company network. The server runs only in a virtual machine on our computer.
- **Domain Name and Public SSL Certificate**: Buying a domain name or a trusted certificate. These cost money and are not needed for a local test setup.
- **Multiple Servers**: Running more than one server or load balancing. Salarize runs on a single server in this activity.
- **New Salarize Features**: Adding automated payroll, attendance tracking, performance reviews, or a mobile app. We only deploy the system as it is.

1.3 Technologies Chosen

Component	Technology chosen	Reason for choice
Web server	IIS (Internet Information Services) 10.0	The built in Windows web server acts as a secure protective layer that guides outside web traffic directly to the Node.js application. This setup keeps the running programs safely separated while strengthening the overall security of the web connections.
Programming Language	Node.js LTS and TypeScript	A fast background system drives the Salarize dashboard by preparing web pages and managing important background tasks. It also carefully restricts access to secure information based on assigned user roles.
Database	PostgreSQL 16	A highly reliable database runs continuously in the background to ensure all company information is stored safely.

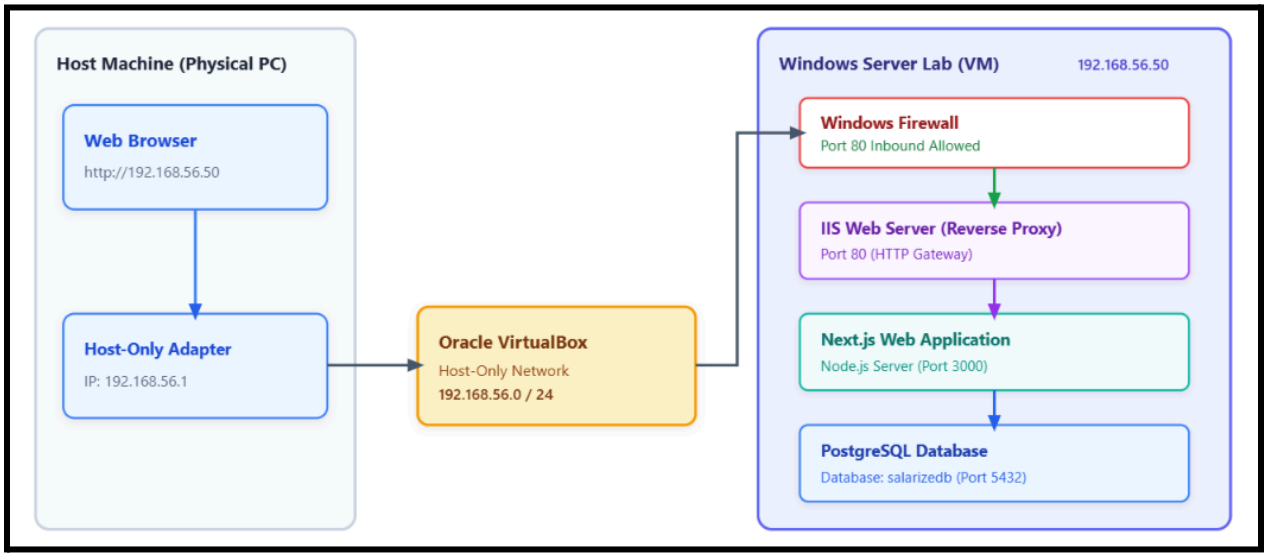
Process Manager	NSSM (Non-Sucking Service Manager)	The main web application is set up to run continuously in the background. It starts automatically whenever the server turns on and will restart itself if any errors occur.
Code Editor and Tools	Visual Studio Code, Oracle VirtualBox, Github	Specialized software creates separate virtual spaces to keep the servers safely isolated from one another. In addition simple command tools are used to easily back up the database. Finally a modern tracking system carefully records any updates made to the project files over time.

EXISTING ENVIRONMENT (BASELINE)

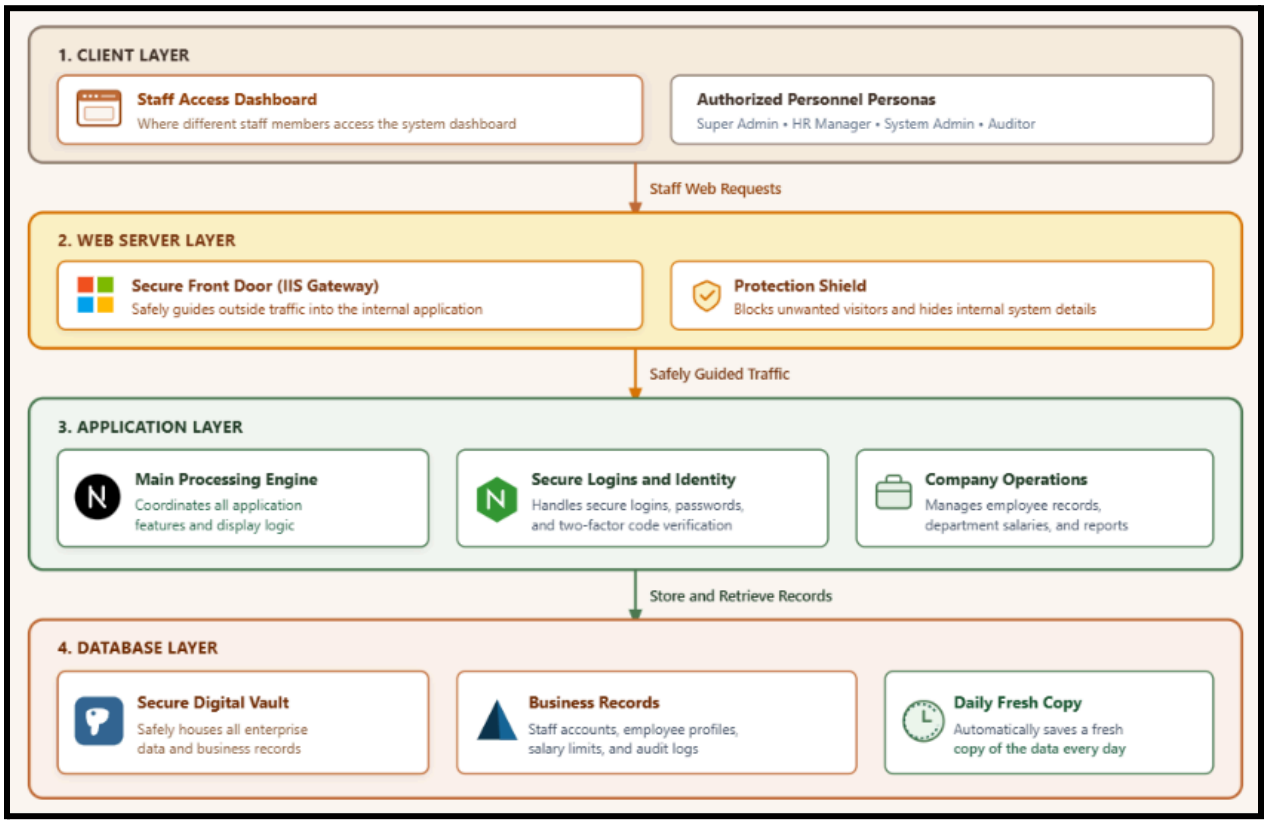
2.1 VM and Server Specifications

Item	Value	Command Used
Windows Server Version and Edition	Windows Server 2022 Standard	winver
Server (Computer) Name	SALARIZE-SRV01	systeminfo
CPU Cores	2 vCPUs	System → Processor
RAM (GB)	4.00 GB (4096 MB)	systeminfo
Disk Size (GB)	50.00 GB	diskmgmt.msc
VirtualBox Network Mode	Host-Only Adapter	VirtualBox VM Settings → Network → Adapter 1
IP Address, Subnet, Gateway, DNS	192.168.56.50	ipconfig /all
Subnet Mask	255.255.255.0	
Default Gateway	192.168.56.1	
DNS	1.1.1.1 or 8.8.8.8	
Baseline Snapshot Name and Date	Snapshot 1: Salarize Initial Testing (Date: October 26, 2026)	Oracle VirtualBox Manager → Snapshots Tab

2.2 Network Diagram



2.3 Architecture Diagram



2.4 Connectivity Check

WEB SERVER SETUP (IIS)

3.1 IIS Installation

3.2 Runtime Installation

3.3 Site Creation

3.4 Verification

DATABASE SETUP

4.1 Installation

4.2 Database

4.3 Schema or ER Diagram

4.4 Sample Data

4.5 User and Permissions

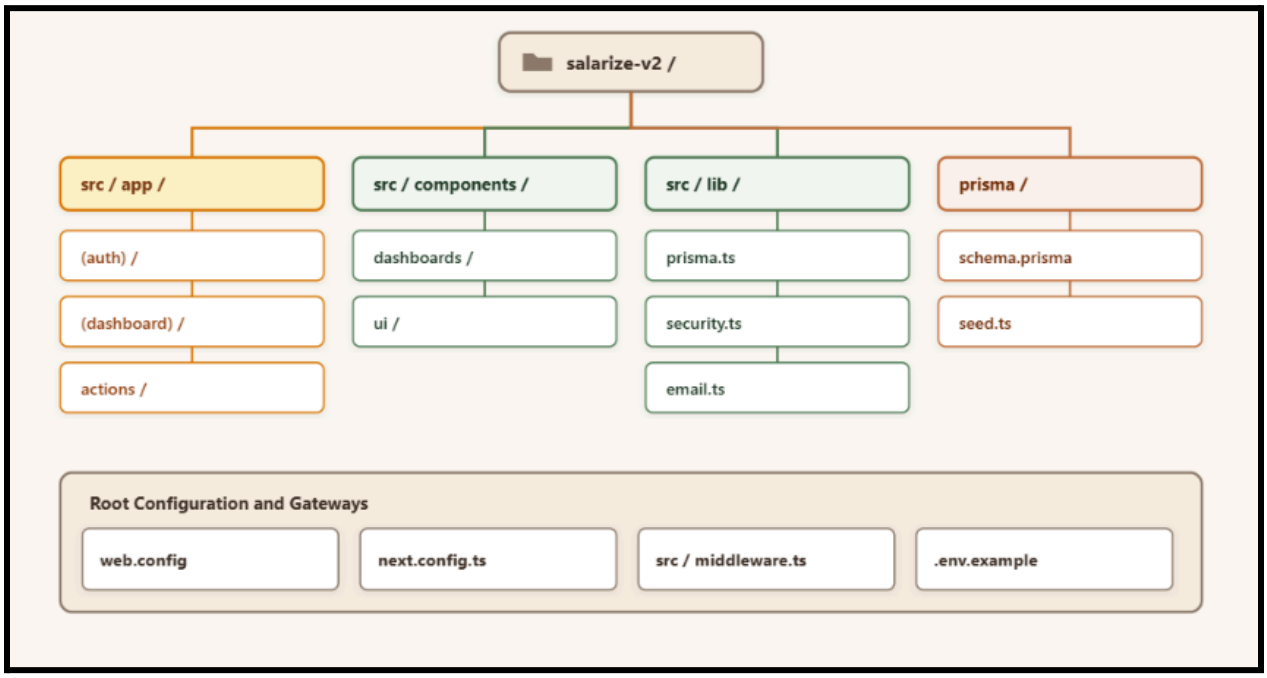
Table	Columns (name, type, key)	Purpose

WEBSITE DEVELOPMENT AND DEPLOYMENT

5.1 Application Overview

Salarize V2 is a complete software platform designed to manage employee records and company payroll. The system strictly limits what each person can access based on their assigned job role. It requires additional security checks whenever a user attempts to view highly sensitive information. Furthermore the application guarantees that every data update is fully completed so no partial or broken records are ever saved.

5.2 File and Folder Structure



5.3 Database Connection

5.4 CRUD Features

5.5 Deployment Steps

5.6 Live Results

Feature	Page / file	Database table used
Create (insert form)		
Read (display records)		

Update		
Delete		

SECURITY CONFIGURATION

Area	What was configured	Reason
Windows Firewall	Port 80 is open for web traffic and port 5432 is locked to the local server.	This ensures outside devices can only reach the main website while keeping the database completely hidden and secure.
User Accounts	The main administrator account has a new name. The standard guest account is turned off and a dedicated account called SalarizeAdmin is set up.	These changes stop automated programs from guessing common login names to break into the system.
Password Policy	Passwords need at least 12 mixed characters. Accounts lock for 15 minutes after consecutive wrong login attempts.	This rule stops hackers from forcing their way in by repeatedly guessing passwords or using lists of common words.
Database User	The specific application user can only open the company database. Remote access for the main system controller is completely blocked.	This gives users only the exact access they need which prevents unapproved people from making major database changes.
Web Server	Folder browsing is turned off. The server uses custom error pages and hides its software version details from public view.	These steps prevent outsiders from gathering helpful clues about how the server is built and operated.
Updates	The system automatically checks for and installs security patches. The software building blocks are also regularly scanned for safety flaws.	This keeps the entire system protected against newly discovered threats and known security weaknesses

TESTING

No.	Feature	Steps	Expected result	Actual result	Pass or Fail
1	Site loads from host machine	Open host Chrome, type http://192.168.56.50	Login page loads cleanly with CSS and brand graphics	Loaded login page with full styling	PASS

2	Insert a record using the form	Log in as HR Manager, go to Employees, click Add Employee, fill details, submit	New employee record saved and the audit ledger creates entry	Employee created with ID, and the audit log recorded	PASS
3	Display records from the database	Navigate to Employee Registry and Personnel tables	All active employee records load in the table	Records populated correctly from PostgreSQL	PASS
4	Update a record	Select an employee, edit salary and position, click Save	Employee profile and salary updated in database	Updated values reflected immediately	PASS
5	Delete a record	Click Terminate Employee on an active staff profile	Status changes to Inactive and the action is logged	Employee marked Inactive and archived	PASS
6	Access from another device (if possible)	From host terminal, execute curl -I http://192.168.56.50	HTTP/1.1 200 OK returned through Host-Only network	HTTP 200 OK received	PASS
7	Database service restarts after VM reboot	Restart Windows Server VM (shutdown /r /t 0), wait for boot, re-test site	PostgreSQL and NSSM services start automatically.	Both services active and the website is accessible immediately	PASS

TROUBLESHOOTING LOG

No.	Problem	Cause	Solution	Screenshot
1	HTTP Error 502.3 - Bad Gateway: The connection with the server was terminated abnormally	The server routing tool had the proxy setting turned off by default	We opened the server manager and checked the box to enable the proxy	
2	Can't reach database server at localhost:5432 during initial Next.js startup	The database was not fully loaded yet when the main application tried to connect.	We added a rule so the main application waits for the database to completely start first	
3	HTTP Error 404.0 - Not Found when requesting static Next.js assets (/_next/static/*)	The server was looking in the wrong place because the visual files were	We copied the missing files into the correct folder and made sure the server	

		saved outside the main web folder.	had permission to read them.	

BACKUP AND MAINTENANCE PLAN

Item	Method used	Frequency	Evidence
VM Snapshots	VirtualBox snapshots taken before each major installation and configuration change.	Before major updates and weekly.	AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA SCREEEENNNSHOOOOTTTT
Database Backup	Automated batch script calling **pg_dump.exe -U salarize_user -d salarizedb -F c -f C:\backups\salarize_%date%.dump**	Daily (6-hour intervals)	AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA SCREEEENNNSHOOOOTTTT
Website Files Backup	Compressed zip archive of C:\Salarize stored in off-site secondary disk directory.	Before major updates and weekly.	AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA SCREEEENNNSHOOOOTTTT
Updates	Regular Windows Server Cumulative Updates and Node.js dependency vulnerability remediation.	Monthly	AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA SCREEEENNNSHOOOOTTTT

CONCLUSION AND REFLECTION

10.1 Conclusion

10.2 Reflection

APPENDICES

Appendix A: SQL Scripts

A.1 Create Database

A.2 Tables

A.3 User

A.4 Sample Data

Appendix B: Configuration Files

B.1 web.config

B.2 php.ini

B.3 Connection Settings with Passwords Hidden

Appendix C: Source Code

Appendix D: Additional Screenshots

REFERENCES